

Romain Giraud

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Senior Software Engineer specialized in **modern C++** and **performance-critical systems**, with additional experience in **Python** and **applied machine learning**. Used to working in R&D environments, driving software from PoC to production, and delivering highly optimized, reliable systems in telecom and avionics contexts, spanning low-level embedded development, backend services, and ML-driven solutions.

Experience

Senior Software Engineer

Oct. 2020 – May 2025

FraudBuster

Marseille, France

- Designed and implemented a **real-time 4G traffic analysis system** capable of processing high-volume network data, using **C** and **DPDK** for low-latency packet handling and a **Python/Django backend** with **Celery**, and **PostgreSQL**, enabling scalable deep packet inspection in a telecom environment.
- Designed and implemented a **proof of concept ML pipeline** to detect fraudulent calls on a GSM network, combining **Go-based network capture** and **audio processing** with **PyTorch models** to generate decision-support indicators for fraud analysts. Evaluated multiple detection approaches, including statistical methods, speech recognition, spectrogram-based classification, and deep learning.
- Joined a newly formed development team following a company merger, taking part in the design and implementation of a Mobile Money product by **building data collection and transformation tools for mobile money transactions**, enabling analysts to efficiently monitor and investigate financial activity.
- Fine-tuned and deployed **machine learning models** to classify scam messages in a Mobile Money fraud context, providing **decision-support indicators** to assist fraud analysts.

Paternity leave

Jan. 2020 – Sept. 2020

World travel

Sept. 2018 – June 2019

Lead Software Engineer

June 2014 – June 2018

Nex Vision

Marseille, France

- Designed and implemented an embedded video surveillance system for the **Airbus A380**, enabling **real-time display of multiple camera feeds** on a single screen, managing video data flow and integration with the Cabin Management System, using **GStreamer** on a Texas Instruments chipset.
- Designed and architected an **embedded gyrostabilized camera system for helicopter applications**, combining a custom PCB with an **NVIDIA Tegra**, **real-time FPGA processing**, and a **C++/Qt/OpenGL software stack**, including low-level Linux driver adaptation for motor and camera control.
- Took ownership of **software architecture definition**, **CI implementation**, and **technical issue resolution**, providing technical leadership across multiple development initiatives.

Freelance

May 2012 – May 2014

Independent

Marseille, France

- Co-created a 2D platformer video game, initially developed in **XNA** and later migrated to **Unity**, redesigning the architecture to support the engine transition. Implemented **interactive environment systems** (water and lava simulation), object-based interactions, and item properties (e.g., wood, stone) to enable gameplay mechanics and puzzle-solving.

Lead Software Engineer

Nov. 2011 – April 2012

So Touch

Marseille, France

- Designed and implemented an interactive **Unity showroom application** to showcase search engine capabilities, enabling users to explore travel destinations on a 3D globe. Integrated **Kinect**, **Leap Motion**, and **3D display support**, including a custom UI system optimized for gesture-based interaction.

Certifications

Certified Kubernetes Application Developer

Online

Linux Foundation

Dec. 2024

Education

Computer engineering degree

Sophia Antipolis, France

Polytech'Nice

Aug. 2008 – Aug. 2011

Interests

Travel, boardgames, sci-fi novels, rock climbing, running, tech-savvy.